

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

Endowing large language models (LLMs) with the capacity to actively retrieve external knowledge is a prerequisite for solving complex, real-world tasks whose answers exceed the model’s parametric knowledge. Existing methods fall into two categories, each with a distinct failure mode: prompting-based approaches merely elicit the model’s latent agentic behavior without optimizing it, whereas reinforcement learning (RL) approaches applied to complex interactive tasks exhibit bounded performance ceilings and training collapse, leaving their agentic capacity underexploited. We address these failure modes with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm engineered for robust agent training via two mechanisms: optimization at the granularity of the full trajectory (sequence-level) and dynamic filtering of rollout samples. The policy is trained end-to-end through RL to interleave multi-turn retrieval and reasoning, eliminating any dependence on supervised demonstration trajectories. Evaluated across multiple question-answering (QA) benchmarks, our 7B-parameter model attains a **34.1%** relative improvement over a comparable prior system, and on complex multi-hop QA such as HotpotQA exceeds the 14B-parameter model of prior work by nearly **9% relative**, while sustaining a stable training trajectory throughout optimization.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) attain strong performance across a range of specialized domains, including mathematical reasoning (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), code generation (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). A structural limitation nonetheless persists: the knowledge encoded in their parameters is static, bounded by the distribution of their pretraining corpus and fixed at its temporal cutoff. A prevailing strategy for circumventing this cutoff is to augment LLMs with a retrieval capability, converting them into agents that issue queries against external knowledge sources at inference time (Jin et al., 2025b). This capability instantiates the more general paradigm of tool use (Schick et al., 2023), in which the model learns a control policy for invoking an external search interface to resolve queries it cannot answer from parametric knowledge alone. Acquiring this competence entails learning a multi-step control policy, which naturally casts the problem as sequential decision-making and thus as an instance amenable to Reinforcement Learning (RL).

In contrast to Supervised Fine-Tuning (SFT), which requires costly static demonstrations and provides no mechanism for exploration, RL admits policy learning from reward signals obtained through trial-and-error interaction (Chu et al., 2025). Within this setting, value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) have become the dominant paradigm, owing to their algorithmic simplicity and their elimination of the memory overhead of a learned critic. Yet, while GRPO performs well on constrained tasks, its application to the open-ended interactive-search domain exposes critical instabilities (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). We attribute this fragility to two distinct sources. First, as characterized by Zheng et al. (2025), GRPO’s token-level surrogate objective is mismatched with a reward that is defined only at the sequence level; the resulting per-token importance weighting induces high-variance gradient estimates that destabilize optimization. Second, the sparse terminal rewards intrinsic to search tasks frequently produce sampled groups with homogeneous outcomes (uniformly successful or uniformly unsuccessful trajectories), which drives the group-normalized advantage to zero and yields a vanishing learning signal, substantially degrading sample efficiency (Yu et al., 2025; Liu et al., 2025).

To resolve these two failure modes—optimization instability and vanishing learning signal—we introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, which composes two complementary mechanisms drawn from recent policy-optimization research. First, DSPO adopts the sequence-level optimization of GSPO (Zheng et al., 2025),

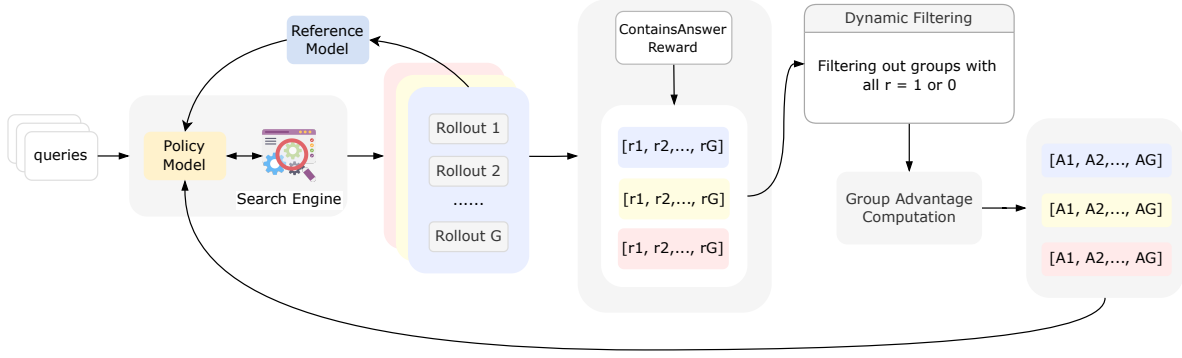


Figure 1: Overview of the DSPO training loop. For a given query, the policy model samples a group of G trajectories through interaction with the search environment, and each trajectory receives a sparse terminal reward. The **dynamic filter** discards groups with homogeneous outcomes and continues sampling until the batch is populated, guaranteeing that every training batch admits a non-degenerate advantage signal. Group-normalized advantages are then computed and applied to update the policy under the sequence-level objective.

defining the importance ratio and the clipped objective over entire trajectories so that the granularity of optimization coincides with the granularity at which reward is assigned; this alignment is what stabilizes optimization over long-horizon reasoning trajectories. Second, DSPO incorporates a dynamic outcome-based filtering mechanism adapted from DAPO (Yu et al., 2025), which constructs each training batch exclusively from rollout groups exhibiting mixed outcomes—at least one successful and one unsuccessful trajectory per prompt—thereby guaranteeing a non-degenerate advantage estimate $\hat{A}_i$. Integrating these two mechanisms within a single framework yields a stable, high-performance optimizer targeted at complex, multi-turn search-and-reasoning tasks. Empirically, our model attains a **34.1% relative improvement** over a leading 7B baseline (Jin et al., 2025b) and, on complex multi-hop benchmarks such as HotpotQA, surpasses even the 14B counterpart (Jin et al., 2025a) by **nearly 9% relative** (0.613 vs. 0.563).

We summarize our contributions as follows:

- We propose DSPO, an RL algorithm that jointly addresses optimization instability and sample inefficiency in the training of agentic search models. DSPO unifies two mechanisms within a single framework: **sequence-level optimization**, which yields low-variance, stable policy updates, and **dynamic outcome-based filtering**, which enforces a dense, non-degenerate learning signal.
- We quantify DSPO’s performance gains under a controlled benchmarking protocol. Our 7B model realizes a **34.1% relative improvement** over a size-matched 7B baseline and, more notably, surpasses the 14B counterpart on complex multi-hop QA, achieving a nearly **9% relative gain** on HotpotQA (0.613 vs. 0.563).
- We provide empirical evidence for DSPO’s training stability, demonstrating a monotonic learning trajectory free of collapse. These results are obtained using only a standard BM25 retriever, which isolates the observed gains as attributable to the optimization algorithm rather than to retrieval quality.

2 RELATED WORK

2.1 RL FOR LLMs

RL for LLMs has progressed through several distinct methodological stages. Early Reinforcement Learning from Human Feedback (RLHF) pipelines couple Proximal Policy Optimization (PPO) with an explicitly trained reward model (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017); subsequent work replaces this two-stage pipeline with direct-optimization frameworks such as DPO (Rafailov et al., 2023). A further shift toward value-free optimization is marked by GRPO (Shao et al., 2024), which dispenses with a learned critic by deriving the advantage from group-level reward statistics. GRPO’s token-level objective, however, is a documented source of training instability (Liu et al., 2025; Cui et al., 2025), which has motivated a series of targeted refinements. GSPO reformulates the objective at the sequence level to match the granularity of the reward (Zheng et al., 2025); DAPO addresses inefficient learning under sparse rewards through a dynamic outcome-based sampling scheme (Yu et al., 2025); and GMPO stabilizes the token-level objective by aggregating token ratios via their geometric mean, reducing

sensitivity to outlier tokens (Zhao et al., 2025). In our experiments, each of these algorithms nonetheless exhibits either training collapse or a performance bottleneck. Building on this line of work, DSPO composes sequence-level optimization with dynamic outcome-based filtering into a single algorithm designed to meet the specific demands of training autonomous search agents.

2.2 LLMs WITH AGENTIC RETRIEVAL

To compensate for the static parametric knowledge of LLMs, Retrieval-Augmented Generation (RAG) couples the model with an external retriever that supplies up-to-date evidence at inference time (Lewis et al., 2020; Gao et al., 2023). Canonical RAG pipelines apply a dense retriever to select relevant documents and concatenate them into the model’s input context prior to generation (Karpukhin et al., 2020). Such pipelines are, however, largely fixed and non-adaptive, which constrains autonomy in complex, multi-turn settings. More recent work moves toward agentic formulations, in which the LLM operates as an autonomous agent that iteratively plans, retrieves, and reasons. ReAct interleaves reasoning traces with actions, enabling tool interaction for tasks such as web navigation (Yao et al., 2023), whereas multi-agent systems such as AutoGen coordinate collaborative workflows across specialized agents (Wu et al., 2024). A parallel direction integrates agentic retrieval with RL, so that retrieval is governed by learned decision-making: Wu et al. (2025) propose Agentic Reasoning, which integrates external tools to streamline LLM inference, while several RL-based methods (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) train LLMs to interleave reasoning and retrieval end-to-end through RL alone. This end-to-end formulation internalizes the agentic control policy within the model and avoids the engineering overhead of orchestrating multiple agents, but it remains susceptible to training instability and to performance ceilings in the open-ended search domain. Our work targets precisely these limitations: DSPO furnishes a robust and sample-efficient training procedure that enforces stable policy optimization and enables LLMs to acquire effective multi-turn search strategies.

3 METHODOLOGY

We proceed in three steps. We first formalize agentic search as an RL problem and review prior policy-optimization algorithms, making explicit the limitations that arise in this regime. We then present **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, specifying the two components responsible for optimization stability and sample efficiency, respectively. Finally, we state the integrated training procedure.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL fine-tuning of LLMs is commonly instantiated with policy-gradient methods, of which PPO (Schulman et al., 2017) is the standard choice for alignment. PPO optimizes a policy π_θ by maximizing a clipped surrogate objective evaluated on trajectories sampled from a behavior (old) policy $\pi_{\theta_{\text{old}}}$. Averaged over tokens, the objective is

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO, however, requires a separately trained value network to estimate the token-level advantages $\hat{A}_t$ via Generalized Advantage Estimation (GAE) (Schulman et al., 2015); this critic incurs substantial memory overhead and constitutes an additional source of optimization instability.

GRPO (Shao et al., 2024) was introduced to remove this dependence on a learned value network. For a prompt x , GRPO samples a group of G responses $\{y_i\}_{i=1}^G$ and assigns each response an advantage by normalizing its reward against the empirical statistics of the group. As in PPO, the surrogate is optimized at the token level:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x,y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x,y_{i,<t})}$, and the advantage assigned to every token $y_{i,t}$ within a response y_i is the common sequence-level value

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

Thus all tokens within a response y_i share a single advantage $\hat{A}_i$ derived from the sequence-level reward, even though the surrogate itself is aggregated token by token—a granularity mismatch we return to below.

Agentic Search as a Markov Decision Process. We formalize the iterative process of agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP), specified by the following components.

- **State (s_t):** At turn t , the state s_t is the concatenation of the full interaction history: the initial question q , all reasoning segments and search queries emitted up to turn t , and the evidence returned by the environment in response to those queries. Since each retrieval result must be parsed and integrated into subsequent decisions, $|s_t|$ grows monotonically along the trajectory, inducing long-horizon dependencies that token-level RL optimizes poorly.
- **Action (a_t):** An action a_t is a contiguous textual segment produced by the policy π_θ , comprising free-form reasoning followed by a terminal decision. The segment terminates in one of two ways: a `</tool_call>` token, which dispatches a search query to the environment and returns its results delimited by `</tool_response>`, or an `</answer>` token, which terminates the trajectory. This structured action space requires the agent to learn jointly *what* content to generate and *when* to retrieve, and consequently trajectory length varies substantially across episodes.
- **Policy (π_θ):** The policy π_θ is the LLM itself, which generates tokens autoregressively conditioned on the current state. Although π_θ conditions on the entire history, the training loss is computed only over model-generated reasoning and action tokens; retrieved evidence tokens are masked out of the objective (Jin et al., 2025b). Optimization therefore demands credit assignment across long sequences in which most intermediate steps receive no direct reward, which reinforces the case for a stable sequence-level objective.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records the complete sequence of reasoning steps and tool interactions for a question, encompassing both model-generated actions and environment-returned evidence. Because the reward is assigned to τ as a whole, the optimization problem is intrinsically sequence-level: any early action can affect terminal answer correctness.
- **Reward ($R(\tau)$):** We adopt a sparse terminal reward that assigns $R = 1$ when the final answer contains the ground-truth text and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we treat the search engine $\mathcal{S}$ as an explicit component of the environment. The policy LLM π_θ generates trajectories by interleaving reasoning with calls to $\mathcal{S}$, and the optimization objective is to obtain a policy maximizing the expected terminal reward, regularized by a KL-divergence penalty that bounds deviation from a fixed reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x;\mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x;\mathcal{S}) || \pi_{\text{ref}}(y|x;\mathcal{S})]. \quad (5)$$

Here the notation $y \sim \pi_\theta(\cdot|x;\mathcal{S})$ indicates that the trajectory y is produced by a multi-step generative process coupling the policy’s autoregressive token generation with the evidence returned by the search engine $\mathcal{S}$.

Motivation for DSPO. Although frameworks such as Search-R1 (Jin et al., 2025b) successfully cast agentic search as an RL problem, instantiating conventional algorithms such as PPO or GRPO in this setting encounters two concrete obstacles. First, the open-ended search environment amplifies the instability of token-level optimization: the granularity mismatch between sequence-level reward assignment and token-level surrogate aggregation (Zheng et al., 2025) produces high-variance gradient estimates whose error compounds over long trajectories and frequently culminates in policy collapse. Second, under the sparse binary reward, a large fraction of sampled groups consist entirely of successful or entirely of unsuccessful trajectories; the group-normalized advantage then degenerates and supplies no learning signal, sharply reducing sample efficiency (Yu et al., 2025; Liu et al., 2025). DSPO is constructed to directly neutralize these two failure modes.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO introduces two mechanisms relative to prior methods: (1) it defines the importance ratio and clipped surrogate at the sequence level, aligning the optimization granularity with the trajectory-level reward structure; and (2) it applies a dynamic outcome-based filter that ensures every training batch carries a non-degenerate, high-quality learning signal. The integrated procedure combining these two mechanisms is illustrated in Figure 1.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Following GSPO (Zheng et al., 2025), we replace the high-variance token-level importance ratio with a sequence-level ratio. For a response y_i , the sequence-level importance ratio $s_i(\theta)$ is defined as the geometric mean of its token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

The exponent $1/|y_i|$ performs length normalization: it reduces the variance of the ratio and confines $s_i(\theta)$ to a consistent numerical range independent of $|y_i|$, which is a precondition for the clipping operation to behave uniformly across trajectories of differing length.

Gradient Analysis. A comparison of gradients makes the stability mechanism explicit. In the unclipped token-level GRPO objective, each token’s log-probability gradient is weighted by a distinct, high-variance, token-specific factor $r_{i,t}(\theta)$. In the sequence-level objective, by contrast, the summed log-probability gradient of the entire trajectory is scaled by a single lower-variance factor $s_i(\theta)$:

$$\nabla_\theta J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_\theta J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_\theta \log \pi_\theta(y_{i,t} | \dots) \right] \quad (8)$$

By applying one trajectory-level correction factor rather than $|y_i|$ independent per-token factors, DSPO suppresses the accumulation of token-level noise that destabilizes prior methods, yielding markedly more stable updates. Complementarily, the dynamic filter guarantees a non-degenerate advantage $\hat{A}_i$ by admitting only rollout groups that contain both successful and unsuccessful trajectories, thereby eliminating wasted samples in the sparse-reward regime.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary reward poses a specific difficulty for group-based advantage estimation. When all G responses in a group share the same outcome—uniformly correct ($R = 1$) or uniformly incorrect ($R = 0$)—the group reward has zero variance, so the normalized advantage $\hat{A}_i$ is either exactly zero or undefined. Such groups contribute no useful gradient and consume rollout compute without benefit.

DSPO removes these degenerate groups via a dynamic filter adapted from DAPO (Yu et al., 2025). During sampling, we retain a group only if it exhibits mixed outcomes. Concretely, a group $\{y_i\}_{i=1}^G$ is admitted for training iff its rewards $\{R_i\}_{i=1}^G$ satisfy

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

Condition equation 9 enforces strictly positive reward variance within every retained group, which guarantees a well-defined and non-vanishing advantage. In effect, this selection curates each policy-update batch to contain only informative groups, converting the sparse-reward problem into a dense and sample-efficient learning signal.

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```
1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for
```

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

Composing the two mechanisms yields the DSPO objective. For each admitted group drawn from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the group-relative normalized advantage

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small positive constant added to the denominator for numerical stability. The policy π_{θ} is then updated by maximizing

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio of Eq. 6, i.e., the geometric mean of the token-level ratios. During likelihood computation we mask the loss on all tokens retrieved from the search tool, following Jin et al. (2025b), so that gradients flow only through model-generated tokens; this ensures the policy is optimized to reason *over* external evidence rather than to reproduce it verbatim. The complete procedure is specified in Algorithm 1.

4 EXPERIMENTS

We now report experiments designed to empirically test the effectiveness of DSPO. The evaluation is organized around three claims: that DSPO (1) attains strong accuracy on challenging question-answering benchmarks; (2) exhibits substantially improved training stability, avoiding the catastrophic collapse observed in the baselines; and (3) derives its gains from the joint action of its two components, as established by controlled ablation.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), we employ the prompt template shown in Table 1 to specify the model’s action interface during the search task, comprising the `<think>`, `<tool_call>`, and `<answer>` control tokens.

Benchmarks and Baselines. Our experimental protocol follows that of Search-R1 (Jin et al., 2025b). We train on a composite corpus formed by the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), and we measure generalization on the test splits of seven QA benchmarks spanning single-hop and

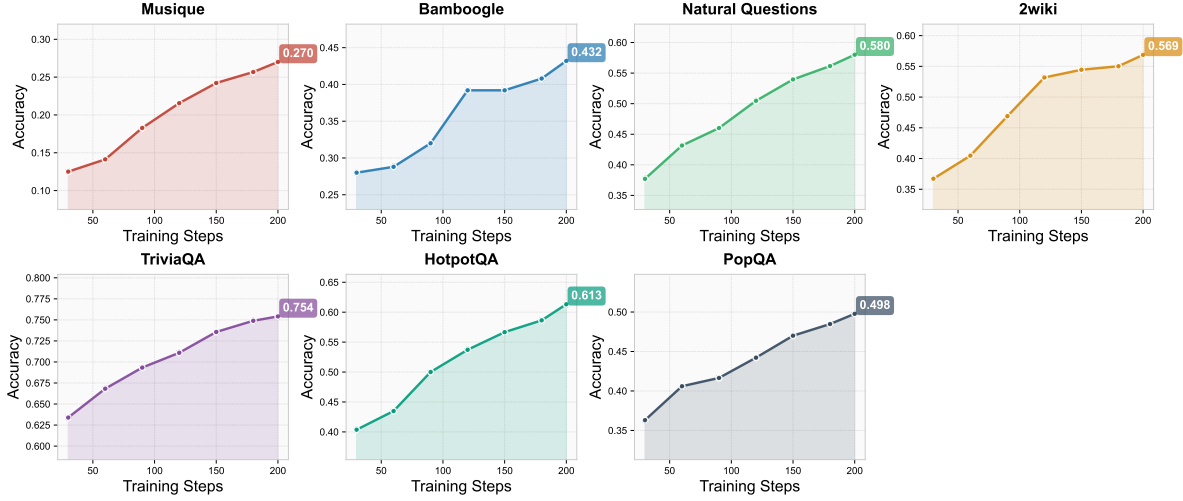


Figure 2: **Validation accuracy of DSPO on seven benchmarks as a function of training progress.** The near-monotonic increase in accuracy indicates that gains in training reward transfer to held-out generalization rather than reflecting reward hacking, evidencing that DSPO converges to a robust search-and-reasoning policy.

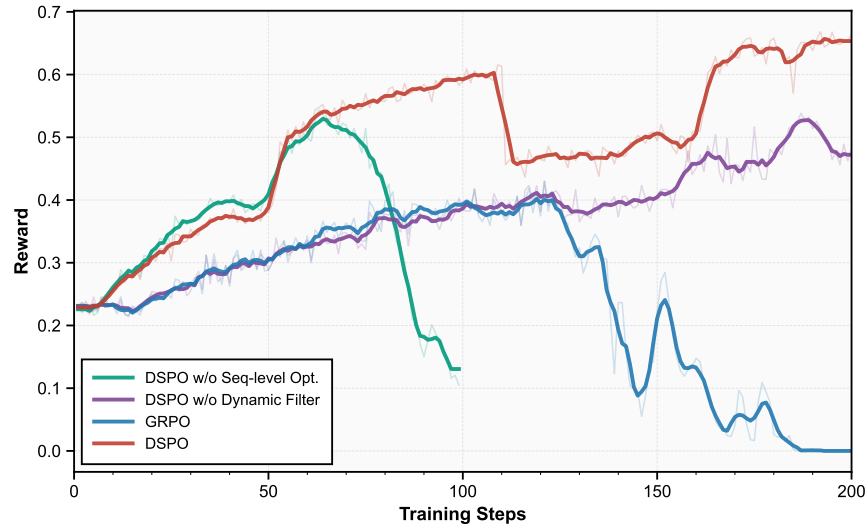


Figure 3: **Training-reward dynamics of DSPO and its ablations.** Learning curves of mean training reward versus training step. DSPO (red) improves stably and near-monotonically; the token-level variants (green, blue) undergo catastrophic policy collapse after a brief initial rise; and the sequence-level variant without the dynamic filter (purple) remains stable but saturates at a suboptimal reward ceiling.

multi-hop settings: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022).

The comparison suite comprises external baselines and internal ablations. The external baselines are the Qwen2.5 7B and 14B models trained with PPO and GRPO within the Search-R1 framework (Jin et al., 2025b;a). To isolate the contribution of each DSPO component, we define two ablations: (1) **DSPO w/o dynamic filter**, which removes the filter and reduces exactly to GSPO (Zheng et al., 2025); and (2) **DSPO w/o sequence-level opt.**, which reverts the objective to the token level, yielding the strong token-level policy DAPO (Yu et al., 2025).

All training runs initialize from the Qwen2.5-7B-Instruct checkpoint. Our implementation extends the VerL framework (Sheng et al., 2025), adapting its provided Search-R1-like example code and training scripts to our method. Because DSPO uses a modified reward function, we do not import previously published numbers directly for

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: Prompt template used to specify the model’s action interface in all experiments.

the external baselines; instead, we retrain the PPO and GRPO models of Search-R1 (Jin et al., 2025b;a) under identical experimental conditions, and these retrained models constitute our primary external reference points.

Implementation and Evaluation. To isolate the contribution of the optimization algorithm from that of retrieval quality, all RL experiments use a fixed, standard BM25 retriever. Under this controlled configuration, any performance difference is attributable to the learned policy rather than to a stronger retriever. Across all methods, the training reward is the sparse binary signal defined by substring Exact Match (subEM) between the final answer and the reference, and subEM likewise serves as the primary evaluation metric.

4.2 MAIN RESULTS AND ABLATION STUDY

Table 2 presents the full comparison. Because DSPO’s two components act jointly, we report the main results and the ablation study in a single table, juxtaposing DSPO against external baselines and its own ablated variants. This layout permits a direct assessment of both overall accuracy and the marginal contribution of each component.

Table 2: Comparison of DSPO against baselines and ablation variants on seven QA benchmarks, reported as subEM. Baselines are the Search-R1 models (7B and 14B) trained with GRPO and PPO (Jin et al., 2025b;a). The ablations remove one component each: ‘w/o dynamic filter’ and ‘w/o seq-level opt.’. Values in parentheses are the originally published Search-R1 EM scores; for consistency, all models reported outside parentheses were retrained and evaluated under our adjusted reward. Best results per row are in **bold** and second-best are underlined.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. Table 2 quantifies DSPO’s advantage. The DSPO-trained 7B agent attains an average subEM of 0.531, the highest among all compared systems, corresponding to a 34.1% relative improvement over the size-matched Search-R1 (GRPO, 7B) model. Notably, this 7B agent marginally exceeds the average of the substantially larger Search-R1 14B models under both GRPO and PPO. The fact that a 7B policy matches a 14B policy indicates that the gains derive from a more effective and stable optimization algorithm rather than from increased parameter count.

Analysis of Ablations. The ablations in Table 2 isolate the marginal contribution of each component. Removing the dynamic filter (‘w/o dynamic filter’) reduces the average subEM to 0.313, a pronounced degradation: absent the filter, the sequence-level objective is repeatedly presented with homogeneous-reward batches and is thus starved of a usable gradient. Reverting the objective to the token level (‘w/o sequence-level opt.’) yields an average subEM of

0.406; although this token-level variant surpasses the filter-less one, it falls well short of full DSPO and, as Section 4 shows in the training-dynamics analysis, is prone to catastrophic instability. Taken together, the two ablations confirm that the components are complementary: sequence-level updates supply optimization stability, while the dynamic filter converts sparse rewards into a dense and sample-efficient learning signal, and neither component alone suffices.

Beyond aggregate metrics, DSPO induces qualitatively sophisticated search behaviors—detection of irrelevant retrieval results, query reformulation, and multi-turn verification (see Appendix A for representative trajectories). These behaviors emerge from RL training alone, without supervised demonstrations.

4.3 ANALYSIS OF TRAINING DYNAMICS

To test the stability and efficiency claims directly, we examine the training-reward trajectories of DSPO, its ablations, and the key baselines. Figure 3 visualizes these trajectories and substantiates the design choices.

DSPO (red) follows a smooth, near-monotonic ascent and converges to the highest reward level, empirically confirming the stability conferred by the sequence-level objective. The **token-level methods**—DSPO w/o seq-level opt. (green) and vanilla GRPO (blue)—instead undergo catastrophic policy collapse early in training: after a brief initial rise, their reward drops sharply, the expected signature of high-variance token-level gradient updates. The **sequence-level variant without the dynamic filter** (purple) remains stable, consistent with sequence-level optimization, but saturates at a markedly suboptimal reward because it lacks a dense learning signal. These trajectories jointly indicate that the combination of sequence-level stability and dynamic filtering is what yields robust and effective optimization; removing either component degrades a distinct axis of performance.

To verify that the reward improvements reflect genuine generalization rather than reward hacking, we track validation accuracy on held-out benchmarks throughout training. As shown in Figure 2, DSPO’s validation accuracy on NQ, HotpotQA, and the remaining benchmarks rises consistently and tracks the training-reward curve. This correspondence between training reward and held-out accuracy indicates that the learned policy generalizes rather than overfitting the reward.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To further characterize the robustness of the method, we extend the evaluation along two axes: scaling to larger model capacity and generalization to a distinct task domain.

Scalability to Larger Models. We test whether DSPO’s stability benefits persist at larger parameter scale by training Qwen2.5-14B-Instruct under identical conditions. As reported in Table 3, the DSPO-trained 14B model attains an average accuracy of 60.6%, exceeding the strong GRPO-14B baseline (53.0%) by a relative margin of **14.3%**. This result indicates that DSPO exploits the additional capacity effectively and that its advantage over standard baselines is preserved—rather than diminished—as model size increases.

Table 3: Scalability analysis on Qwen2.5-14B-Instruct, reported as per-benchmark accuracy. The final column reports the relative gain of DSPO (14B) over the GRPO (14B) baseline. Best results per row are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. To probe whether DSPO’s benefits extend beyond agentic search, we apply it to single-turn mathematical reasoning using the Qwen2.5 and Qwen3 model families. Table 4 reports the comparison on Math500 and Olympiad-Bench, where DSPO exceeds GRPO at both the 7B and 4B scales. These

results suggest that DSPO’s stabilizing and signal-densifying mechanisms are not specific to search and transfer to general reasoning domains.

Table 4: Generalization to single-turn mathematical reasoning, reported as accuracy. The final column reports the relative gain of DSPO over GRPO under a matched training-step budget. Best results per row are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

This work addressed the optimization instability and sample inefficiency that limit RL for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), which enforces robust training through two mechanisms: sequence-level optimization, which aligns the optimization granularity with the trajectory-level reward and thereby prevents catastrophic policy collapse, and a dynamic outcome-based filter, which restricts each batch to mixed-outcome groups and thus converts sparse rewards into a consistently non-degenerate learning signal. Empirically, DSPO both attains strong accuracy across a suite of challenging QA benchmarks and exhibits substantially improved training stability relative to prior methods.

By supporting robust training from environmental reward alone, DSPO offers a practical and reproducible recipe for building capable LLM agents without curated expert demonstrations. Building on this stable foundation, future work may integrate stronger retrievers or extend the method to multi-tool settings. Moreover, since sparse rewards and high-variance policy gradients are not specific to search, we conjecture that DSPO’s mechanisms will confer analogous stability and accuracy gains in adjacent domains such as mathematical and code generation—a hypothesis our preliminary math-reasoning results support and that warrants systematic validation. We expect the two core principles of DSPO—matching the granularity of optimization to that of the reward, and guaranteeing signal density—to be broadly applicable to the design of autonomous agents.

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A APPENDIX

To characterize the learned search strategies qualitatively, Table A.1 presents representative decoded trajectories. These instances illustrate three learned behaviors: (1) query reformulation when an initial retrieval returns irrelevant evidence, (2) verification of a candidate answer through follow-up retrieval, and (3) extraction of a complete answer from the retrieved evidence.

Table A.1: Representative decoded trajectories illustrating the model’s learned search-and-reasoning behavior. The examples exhibit (a) recognition of irrelevant retrieval results, (b) query reformulation, and (c) answer verification through follow-up retrieval.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let's try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin’s occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American <u>journalist, attorney, soldier, and politician</u>. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician